3.0 Technical Description

This chapter describes the AirLume lightning prediction system. Section 3.1 defines the system and its architecture. Section 3.2 explains each major component. Section 3.3 demonstrates one complete operational cycle.

3.1 General Description

3.1.1 Technical Definition

AirLume is a multi-language lightning prediction system for aviation safety. It analyzes real-time atmospheric conditions to forecast lightning formation probability 2-4 hours in advance along flight routes. The system combines plasma physics models with machine learning to generate risk assessments. Results are delivered through an enterprise web interface.

3.1.2 Physical Description

AirLume consists of five software assemblies operating in sequence. The architecture follows ARINC 653 partitioning principles, separating safety-critical components from non-critical layers.

The five assemblies:

- **Python Weather Layer –** Retrieves atmospheric data from OpenWeatherMap API. Extracts temperature, humidity, pressure, and wind measurements. Outputs formatted data to text file.
- **C Physics Engine –** Implements Paschen's Law and Townsend Avalanche algorithms. Calculates electric field strength at cruise altitudes. Generates waypoints along flight routes. Outputs lightning risk percentages.
- **Python ML Enhancement –** Applies trained Random Forest model to physics predictions. Adjusts risk estimates based on historical accuracy patterns. Outputs enhanced predictions with confidence scores.
- **Ada Safety Monitor –** Performs independent risk calculations using simplified algorithms. Compares results against C and ML outputs. Generates SAFE/CAUTION/EMERGENCY alerts when discrepancies exceed thresholds. Maintains audit logs for regulatory compliance.

- **Jakarta EE Web Application –** Orchestrates execution of all components. Parses outputs and stores results in database. Renders interactive visualizations with color-coded risk indicators.

System Requirements: Cloud infrastructure (AWS/Azure), weather API subscriptions (OpenWeatherMap), GlassFlight application server, Python 3.x, GCC C compiler, GNAT Ada compiler, Java JDK 11+, scikit-learn library.

[Figure 1: System Architecture Diagram showing data flow through five assemblies]

3.2 Part-by-Part Description

3.2.1 Python Weather Integration Layer

Purpose: Acquires atmospheric data from external weather services.

Implementation: Standalone Python script (weather_api.py, 250 lines). Uses standard libraries for HTTP requests, JSON parsing, and file operations.

Three sub-components:

API Connection Module

- Establishes HTTPS connections to OpenWeatherMap
- Implements 30-second timeout to prevent hanging
- Uses exponential backoff retry logic (2s, 4s, 8s delays)
- Validates HTTP status codes (200=success, 401=auth error, 429=rate limit, 500=server error)
- Logs all activity to weather_api.log

Data Parser

- Deserializes JSON responses into Python dictionaries
- Extracts temperature (Kelvin), humidity (%), pressure (hPa), wind speed (m/s)
- Validates field presence and physical ranges: Temperature 200-330K, Humidity 0-100%, Pressure 870-1085 hPa, Wind speed 0-150 m/s
- Rejects invalid measurements

Output Formatter

- Writes standardized format: WEATHER_DATA:temp,humidity,pressure,windspeed
- Uses atomic file operations (write temp file, then rename)
- Rounds values to two decimal places
- Example: WEATHER_DATA:268.15,72.0,1018.0,8.5

[Figure 2: Python Weather Layer flowchart]

3.2.2 C Physics Calculation Engine

Purpose: Performs plasma physics computations for lightning prediction.

Implementation: 1,800 lines across modular C files. Core modules: lightning_physics.c (calculations), route.c (waypoint generation), atmosphere.c (ISA corrections), io.c (file operations). Compiles with GCC -O2 optimization.

Four sub-components:

Atmospheric Data Processor

- Reads weather_data.txt and parses comma-separated values
- Applies International Standard Atmosphere (ISA) corrections
- Adjusts ground-level measurements to flight altitudes (20k, 30k, 40k feet)
- Calculates air density for ice crystal concentration modeling

Electric Field Calculator

- Implements Paschen's Law to determine breakdown voltage from pressure and gap distance
- Models charge accumulation from ice crystal collisions in clouds
- Calculates atmospheric electric field strength
- Applies Townsend Avalanche criterion for electron multiplication
- Generates risk ratio: >1.0 (imminent), 0.7-1.0 (elevated), <0.5 (low probability)

Waypoint Generator

- Divides flight route into 8 equal segments (9 waypoints total)
- Uses great circle calculations for shortest path on Earth's surface
- Applies spherical interpolation to compute intermediate coordinates
- Handles special cases: Date Line crossings, polar routes, antipodal points

- Stores results in lat[9] and lon[9] arrays

Risk Assessment Module

- Retrieves weather conditions at each waypoint position
- Feeds atmospheric parameters to Electric Field Calculator
- Converts risk ratios to percentage probabilities using calibrated scaling
- Performs multi-altitude analysis at FL200, FL300, FL400
- Writes results to lightning_risk.txt with detailed calculation logs

[Figure 3: C Physics Engine component diagram]

3.2.3 Python ML Enhancement Layer

Purpose: Enhances physics predictions using machine learning trained on historical data.

Implementation: Python script (ml_enhance.py, 400 lines). Uses scikit-learn Random Forest classifier. Loads pre-trained model from airlume_ml_model.pkl.

Three sub-components:

Feature Extractor

- Reads physics prediction from lightning_risk.txt
- Reads atmospheric data from weather_data.txt
- Extracts features: physics risk, temperature, humidity, pressure, altitude, latitude, longitude
- Normalizes features to consistent scales
- Creates feature vector for model input

ML Predictor

- Loads trained Random Forest classifier (100 estimators)
- Feeds feature vector to model
- Outputs enhanced probability prediction
- Calculates adjustment factor (ML prediction minus physics prediction)
- Provides confidence score based on feature importance

Output Generator

- Writes to lightning_risk_ml.txt
- Format: ML_ENHANCED:XX.XX%, PHYSICS:XX.XX%, ADJUSTMENT:+/-XX.XX%

- Includes confidence level and feature importance rankings
- Logs predictions for performance monitoring

Training Details: Trained on 100 real atmospheric measurements from USA flights (airlume_usa_efield_100.csv). Contains actual electric field data and lightning occurrence labels. Top features by importance: latitude (23%), physics prediction (21%), longitude (14%), humidity (10%). Achieves 45% accuracy, representing 10% improvement over physics-only approach. 5-fold cross-validation shows 45% mean accuracy.

[Figure 4: ML Enhancement Layer workflow]

3.2.4 Ada Safety Validation Monitor

Purpose: Independent verification for safety-critical accuracy and regulatory compliance.

Implementation: 600 lines organized into three Ada packages: Lightning_Physics (calculations), File_IO (operations), Safety_Logger (audit trails). Compiles with GNAT using -gnatwa (all warnings) and -gnatVa (validity checking) flags.

Three sub-components:

Independent Calculator

- Uses empirical risk models based on meteorological observations
- Defines strongly-typed variables with range constraints
- Applies weighted scoring: Risk = (0.3 × temperature) + (0.4 × humidity) + (0.3 × pressure)
- Produces estimates that typically correlate >0.85 with C engine
- Different algorithmic approach enables independent bug detection

Validation Comparator

- Reads C physics output from lightning_risk.txt
- Reads ML enhanced output from lightning_risk_ml.txt
- Executes independent Ada calculations
- Computes absolute differences between all three predictions
- Categorizes: SAFE (<15% difference), CAUTION (15-25%), EMERGENCY (≥25%)
- Logs all comparisons with timestamps for regulatory audits

Safety Alert Generator

- Writes structured alerts to safety_status.txt
- Format includes: SAFETY_STATUS, C_RISK, ML_RISK, ADA_RISK, DIFFERENCE, JUSTIFICATION, TIMESTAMP
- For EMERGENCY alerts: logs to emergency.log and sends email/SMS notifications
- Continues operation after alert generation to allow fault observation
- Architecture supports DO-178C Level A certification requirements

[Figure 5: Ada Safety Monitor architecture]

3.2.5 Jakarta EE Enterprise Web Application

Purpose: User interface for flight planners. Orchestrates all system components and presents results.

Implementation: 2,500 lines of Java on GlassFish 6.2 server. Model-View-Controller architecture with JPA entities, backing beans, and XHTML templates. H2 embedded database for persistence.

Four sub-components:

Process Orchestrator

- Receives user inputs via web forms (departure city, arrival city, cruise altitude)
- Converts airport codes to coordinates using internal database
- Executes components sequentially: Python weather → C physics → Python ML → Ada validation
- Enforces 60-second timeout per component
- Validates coordinates, verifies executable files, confirms output file creation
- Logs all executions to application.log

Output Parser

- Reads text output files from all components
- Applies regular expressions to extract structured data
- Populates Java objects (FlightAnalysis entity, Waypoint list)
- Validates numeric ranges
- Handles parsing errors with default values and warning logs

Data Persistence Layer

- Uses Jakarta Persistence API (JPA) with Hibernate
- FlightAnalysis entity stores route, altitude, risks, timestamp, safety status
- OneToMany relationship to Waypoint records

- H2 database tables: flight_analysis, waypoint
- Indexes on analysis_time and safety_status for query performance

Visualization Renderer

- Generates HTML using JavaServer Faces (JSF) templating
- Creates HTML5 canvas maps with JavaScript
- Draws color-coded waypoint circles: Green (0-25% risk), Yellow (25-50% risk), Red (50-100% risk)
- Connects waypoints with matching color line segments
- Displays data table with coordinates, altitudes, risk percentages
- Shows prominent alert banner for CAUTION/EMERGENCY status
- Provides input forms, altitude selector, progress indicator, analysis history, CSV export

[Figure 6: Jakarta EE Web Application components]

3.3 Cycle of Operation

This section demonstrates one complete operational cycle using a real flight route example.

3.3.1 Initialization and Weather Acquisition

Flight planner accesses AirLume web interface. Selects route from dropdown menus: Ottawa International (CYOW) to Toronto Pearson (CYYZ). Chooses cruise altitude FL300 (30,000 feet). Clicks "Analyze Route" button.

Jakarta EE backing bean converts airport codes to coordinates: CYOW (45.32°N, 75.67°W) and CYYZ (43.68°N, 79.63°W).

Process Orchestrator launches Python weather script. Script requests weather data for 8 waypoints along the route. OpenWeatherMap API responds with current atmospheric conditions. Sample waypoint data: Waypoint 1 (Ottawa) -67.7°C, 49% humidity, 300 hPa, 7.4 m/s wind; Waypoint 4 (midpoint) -63.5°C, 63% humidity, 300 hPa, 6.0 m/s wind; Waypoint 8 (Toronto) -62.2°C, 47% humidity, 300 hPa, 7.7 m/s wind.

All measurements pass validation. Output Formatter writes 8 records to weather_data.txt. Python exits with status 0.

3.3.2 Physics Calculations

C physics engine reads weather data. Atmospheric Data Processor applies ISA corrections for FL300 altitude. Adjusts temperatures, pressures, and calculates air densities.

Waypoint Generator computes great circle route from Ottawa to Toronto. Total distance: 363.5 km. Divides into 8 segments with waypoints at 52 km intervals.

Electric Field Calculator analyzes each waypoint: Waypoint 1 with E-field 219 V/m and low ice crystal density produces 4.2% risk (LOW). Waypoint 3 with E-field 847 V/m and moderate ice density produces 25.8% risk (MODERATE). Waypoint 6 with E-field 848 V/m and high humidity produces 26.0% risk (MODERATE - PEAK). Waypoint 8 with E-field 657 V/m and lower humidity produces 13.7% risk (LOW).

Risk Assessment Module calculates overall route statistics: Maximum risk 26.0% at waypoint 6 (260 km from origin), average risk 18.6%. Writes detailed results to lightning_risk.txt.

3.3.3 ML Enhancement

ML layer reads physics prediction: 26.0% maximum risk. Feature Extractor pulls atmospheric data: Latitude 43.92°N, Temperature -65.2°C, Humidity 64%, Pressure 300 hPa, Physics prediction 26.0%.

ML Predictor loads Random Forest model. Feeds feature vector to classifier. Model processes based on training data patterns. In this case, model agrees with physics calculation.

Output: ML_ENHANCED: 26.0%, PHYSICS: 26.0%, ADJUSTMENT: +0.0%. Writes to lightning_risk_ml.txt. Confidence: HIGH (physics and ML aligned).

3.3.4 Safety Validation

Ada monitor reads predictions: C physics 26.0%, ML enhanced 26.0%. Independent Calculator executes empirical model using weather data. Weighted scoring yields Ada prediction: 23.5%.

Validation Comparator calculates differences: |C - Ada| = 2.5% → VALIDATED (<15% threshold), |ML - Ada| = 2.5% → VALIDATED. Safety status: CAUTION (moderate risk level, not discrepancy-based).

Alert Generator writes to safety_status.txt: SAFETY_STATUS: CAUTION, C_RISK: 26.00%, ML_RISK: 26.00%, ADA_RISK: 23.50%, DIFFERENCE: 2.50%, JUSTIFICATION: Enhanced weather monitoring required. All models show moderate risk.

No emergency notifications triggered. Logs saved for audit trail.

3.3.5 Result Presentation

Output Parser extracts all data: Physics prediction 26.0%, ML enhanced 26.0%, Ada validation 23.5%, Safety status CAUTION, 8 waypoint records with individual risks.

Data Persistence Layer creates database records: FlightAnalysis (route=CYOW→CYYZ, altitude=30000, max_risk=26.0, avg_risk=18.6, status=CAUTION) and 8 Waypoint records with coordinates, altitudes, individual risk percentages.

Visualization Renderer generates HTML results page. Canvas map plots 8 waypoints on route. Color coding: Waypoint 1 green (4.2%), Waypoints 2-6 yellow (16-26%), Waypoints 7-8 green (13.7%). Peak risk highlighted at waypoint 6 (260 km mark). Data table shows complete details. Yellow CAUTION banner displays: "Enhanced monitoring required. Consider altitude change to FL350+".

Browser renders interactive visualization. Total processing time: 22 seconds.

3.3.6 Value to User

The system identified moderate lightning risk along the Ottawa-Toronto route. Peak risk (26%) occurs over the central segment at 260 km from departure. This information enables informed decisions.

Route Optimization Options: Request altitude change to FL350 or FL400 where conditions may be safer. Plan slight route deviation (30-50 miles north or south) around waypoint 6. Delay departure 1-2 hours to allow weather system movement. Proceed as planned with enhanced weather radar monitoring.

Safety Assurance: CAUTION alert confirms moderate risk level. 2.5% difference between models shows excellent agreement. Physics, ML, and Ada predictions aligned. Planner can trust results while maintaining appropriate vigilance.

Operational Benefits: Database persistence enables historical trend analysis. Seasonal pattern identification for this route. Correlation tracking between predictions

and actual outcomes. ML model continuously improves with additional real flight data. Multi-altitude analysis supports fuel efficiency optimization. Reduces costly post-strike inspections through proactive avoidance. Minimizes weather-related delays and diversions.

[Figure 7: Complete Operational Cycle diagram]